p_0

$$E^* = -\frac{2(ke^2)^2mc^2}{(\hbar c)^2} \approx -54.4\ \text{eV} . \qquad (3)$$

E_0 2

$$E_0 = 2E^* \approx -108.8\ \text{eV} . \qquad (4)$$

E^* E_0

$$E^+ = E^* - E_0 = -E^* \approx 54.4\ \text{eV} . \qquad (5)$$

$$E^* = \frac{p^2}{2m} - \frac{2ke^2}{r} .$$

$$rp = \hbar \qquad (3)$$

$$E^* = \frac{\hbar^2}{2mr^2} - \frac{2ke^2}{r} = \frac{\hbar^2}{2m}\left(\frac{1}{r} - \frac{2ke^2m}{\hbar^2}\right)^2 - \frac{2k^2e^4m}{\hbar^2} .$$

（4）

$$r_0 = \frac{\hbar^2}{2ke^2m} , \quad E^* = -\frac{2k^2e^4m}{\hbar^2}$$

（1），（ 2）

（II） E_0 r_0

$$\frac{mv^2}{r} = \frac{2ke^2}{r^2} - \frac{ke^2}{(2r)^2} = \frac{7ke^2}{4r^2}$$

$$rmv = \hbar$$

$$r_0 = 4\hbar^2 / 7ke^2m , \qquad (6)$$

$$E_0 = 2\left(\frac{p_0^2}{2m} - \frac{2ke^2}{r_0}\right) + \frac{ke^2}{2r_0} = -\frac{49k^2e^4m}{16\hbar^2} ; \qquad (7)$$

$$E_0 = -\frac{49(ke^2)^2mc^2}{16(\hbar c)^2} \approx -83.4\ \text{eV} , \qquad (8)$$

$$r_0 = \frac{4(\hbar c)^2}{7ke^2mc^2} \approx 0.0302\ \text{nm} .$$

$$E^+ = E^* - E_0 \approx 29.0\ \text{eV} . \qquad (9)$$

2 4 6 4 .